

Project Abstract: OraTrack

ML Powered Oral Cancer Detection and Smart Oncology Patient Management System

Abstract

Oral cancer continues to be a major public health concern, particularly in regions where early detection is limited due to delayed diagnosis, manual clinical evaluation, and fragmented patient records. Many hospitals still depend on paper-based systems for screening, biopsy processing, and oncology treatment management, leading to slow coordination, miscommunication, and reduced treatment effectiveness. OraTrack addresses these challenges by providing an integrated digital platform that combines ML-powered cancer risk prediction, automated biopsy workflow, and a complete oncology patient management system. The platform uses doctor-entered clinical parameters and patient-entered parameters including lesion size, ulcer duration, pain intensity, bleeding, lifestyle factors, and medical history to generate an accurate cancer risk prediction (Normal, Low, Moderate, or High) using a pre-trained Random Forest model. When the predicted risk is moderate or high, the system automatically creates a biopsy request, assigns the appropriate specialist, and tracks progress through QR-based status updates. The Lab Technician module further strengthens this workflow by allowing technicians to view doctor-assigned biopsy requests, scan QR codes to update each stage (sample collected, lab received, under analysis, report generated), upload biopsy findings with digital pathology reports, notify doctors automatically when results are ready, and maintain full transparency by recording lab processing time. Once a biopsy confirms cancer, the system automatically transfers the case to an oncologist, along with all previous clinical details, predictions, and symptoms. The oncology module supports treatment planning, therapy scheduling, symptom monitoring, and progress evaluation. Patients can submit daily symptoms, therapy side effects, and medication updates, receive reminders, and interact directly with doctors through a two-way communication channel. The system also supports emergency alerts, allowing patients to request immediate help during critical situations. By integrating AI-assisted screening, transparent biopsy tracking, specialist reassignment, continuous symptom monitoring, emergency support, and structured oncology management, OraTrack ensures faster diagnosis, reduces workflow delays, strengthens doctor–patient communication, and delivers a seamless digital pathway from initial screening to long-term cancer care.

User Modules

1. Admin

- Add and manage hospital departments, doctors, and support staff.
- Register and manage patients, doctors, and system users.
- View and monitor all screening submissions and ML predictions.
- Oversee biopsy workflow and verify uploaded pathology reports.
- Manage oncology treatment schedules, appointments, and follow-up planning.
- Generate analytical reports: screening statistics, biopsy turnaround time, treatment history, doctor performance, etc.
- Handle patient complaints, system feedback, and notifications.

- Maintain security, role permissions, and system configuration.
- Receive and respond to emergency help alerts sent by patients during critical health conditions.
- Monitor two-way patient–doctor interactions, including messages, alerts, and follow-up guidance.

2. Doctor / Oncologist

- Enter clinical parameters such as lesion appearance, ulcer duration, lifestyle habits, symptoms, and patient history.
- Submit values to the ML model and receive risk prediction (Normal / Low / Moderate / High).
- Approve or generate biopsy requests based on prediction or clinical judgment.
- Track biopsy status through QR-based updates within the system.
- Access and review biopsy reports and record final diagnosis.
- Create personalized treatment plans for cancer patients.
- Schedule follow-up visits, chemotherapy/radiotherapy sessions, and required medical tests.
- Monitor ongoing patient symptoms and treatment response submitted by the patient.
- Send alerts, advice, and instructions directly to the patient.
- View patient progress analytics and recurrence-risk insights.
- Receive emergency help requests from patients and provide immediate assistance.
- Engage in two-way communication, allowing both patient-to-doctor and doctor-to-patient messaging.
- Automatically take over the treatment workflow when a biopsy confirms cancer—the system auto-assigns an oncologist and transfers all previous data (screening values, predictions, symptoms, reports) from the initial doctor.
- Continue and manage the remaining oncology treatment from the point of diagnosis

3. Patient

- Enter basic health details, symptoms, and lifestyle information.
- View ML-based screening results sent by the doctor.
- Receive notifications for screening results, biopsy request, biopsy status, and appointment reminders.
- View QR code to check biopsy progress (Requested → Under Review → Report Ready).
- View biopsy reports, prescriptions, treatment plans, and upcoming appointments.
- Submit daily symptoms and medication updates for continuous monitoring.
- Access complete treatment history, doctor recommendations, and visit summaries.
- Submit complaints or request assistance through the portal.
- Send emergency help requests and critical alerts to both doctors and administrators.
- Participate in a two-way communication system for messaging, advice, alerts, and follow-up support.
- At first appointment, choose a doctor from General Dentist / Oral Health Specialist / ENT based on preference.

- After biopsy request, the system automatically assigns an oncologist, and the patient's treatment flow moves to that doctor.
- Submit daily post-treatment symptoms and therapy side effects for real-time monitoring.

4. Lab Technician

- View biopsy requests assigned by doctors.
- Scan QR codes to update biopsy status at each stage: sample collected → lab received → under analysis → report generated.
- Upload biopsy findings and digitally attach pathology reports.
- Notify doctors automatically when reports are ready.
- Record lab processing time and maintain transparency in workflow.

Tools Used

- **Python:** Backend development and machine learning processing.
- **Django:** Web application framework for server-side operations.
- **Scikit-learn:** Used to train the Random Forest model for cancer risk prediction.
- **Pandas & NumPy:** Data handling and preprocessing.
- **MySQL:** Database for storing users, clinical data, predictions, and reports.
- **HTML, CSS, JavaScript:** Frontend interface design.
- **Bootstrap:** For responsive and user-friendly UI.
- **QR Code Library:** Generates QR codes for biopsy tracking.